Demand model

`noga/model.py @ 96709cd+dirty`

not inputs: $p_{da}(t)$, $p_o(t)$, $p_u(t)$

Demand history
same slot, last 7 days, MW
$t-7\text{d}$ … $t-1\text{d}$

Calendar
month (12) · weekday (7)
5-min slot of day (288)

3 indices

Embeddings
`Embedding(12, 2)` → 2
`Embedding(7, 2)` → 2
`Embedding(288, 2)` → 2
each vector: norm ≤ 1

6

Weather forecast for t
3 vars × 3 stations = 9
temp, wet bulb, humidity

9

Standardise
weather + $p_{da}(t-24\text{h})$
$(x-\mu)/\sigma$ per column
train μ, σ

10

network input: calendar, weather, and $p_{da}(t-24\text{h})$ only

Day-ahead price history
4 prices 30 min apart, ILS/MWh
$t-24\text{h}-1.5\text{h}$ … $t-24\text{h}$

$p_{da}(t-24\text{h})$, 1

Concatenate
2 + 2 + 2 + 9 + 1 = 16

16

Linear
16 → 16

16

repeated block

LayerNorm
16

LeakyReLU
→ **Dropout** 0.1

Linear
16 → 16

× 2

16

LeakyReLU

Shared features z
16

16

Demand head: Linear
16 → 9

16

Price head: Linear
16 → 7

cost model only

7

Attention logits
7 → softmax
→ $w_1 \dots w_7$

2

Scale and shift
$a = e^s > 0$, starts at 1
$b = \tilde{b} \cdot y_{\text{scale}}$

4

Attention logits
4 → softmax
→ $v_1 \dots v_4$

3

Scale factors
s_{da}, s_o, s_u (3)

7

Attention
$\text{att} = \sum_i w_i \cdot \text{hist}_i$

4

Attention
$\text{att}_p = \sum_j v_j \cdot \text{price}_j$

Demand forecast
$\hat{y} = a \cdot \text{att} + b$ (MW)

Price forecast (ILS/MWh)
$\hat{p}_{da} = e^{s_{da}}\,\text{att}_p$ (starts at ×1)
$\hat{p}_o = e^{s_o}\,\hat{p}_{da}$, $\hat{p}_u = e^{s_u}\,\hat{p}_o$

Training

Demand loss (one per model, on $\hat{y}$)

cost (proposed): mean excess cost
$(p_u - p_{da})\,(y-\hat{y})^+ + p_o\,(\hat{y}-y)^+$

baselines: L2, L1, Pinball med / opt / p95
(constant quantile from train prices)

losses use $\hat{y}/y_{\text{scale}}$ and p/p_{scale} (train split):
y_{scale} = std of $y(t)-y(t-24\text{h})$, p_{scale} = mean p_{da}

Price loss (cost model only)

relative squared error, real prices:
mean$\left[((\hat{p}-p)/p)^2\right]$ over p_{da}, p_o, p_u
total loss = demand loss + 1.0× price loss

Optimiser

Adam, lr 10^{-3}, no weight decay; dropout 0.1 in the trunk
300 epochs, fixed (no early stopping)
batches of 5 000 slots, reshuffled every epoch
train 2023-04 … 2024-12, test 2025

inputs · preprocessing · shared trunk · demand head · price head · training only

→ raw history (≥ 24 h old) into an attention (only $p_{da}(t-24\text{h})$ also enters the network)